

Henry Matar

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Education

University of Colorado Boulder, College of Engineering and Applied Science

Expected May 2027

B.S. in Electrical Engineering

Minors in Computer Science & Applied Mathematics

- **GPA: 3.98/4.00** | Dean's List: All Eligible Semesters
- Tau Beta Pi | Engineering Honors Society

Skills

Software: Altium Designer, Linux, C, C++, Java, Git, GitHub, GitLab, JIRA, Verilog, MATLAB, Python, MicroPython, SPICE (SIMetrix, LTSpice), Signal Processing, STM32, CAD (Onshape)

Hardware: PCB Design, Oscilloscopes, Soldering, RF Systems, Spectrum Analyzers, Embedded Systems, FPGA Development

Work Experience

BAE Systems / CU Boulder Capstone Design Project – *Technical Lead, Post-Processing*

Aug 2026 – Present

- Developing a bistatic drone-based RF radar system with 2D imaging, object localization, and scattering characterization.
- Inherited a loopback-validated radar signal-processing codebase and led its transition to outdoor free-space testing, raising TX/RX gain and pulse count to lift SNR from ~5 to 15–25 dB above the noise floor, enabling reliable wall-distance estimates.
- Replaced the rectangular chirp window with a Kaiser window ($\beta = 10$), cutting window side lobes from -13 dB to below -70 dB so weak target returns remain detectable beside the strong direct-path signal.

Lockheed Martin, Systems Engineering Intern – Mount Laurel, NJ

Jun 2026 – Aug 2026

- Automated end-to-end messaging validation by scripting REST API-driven tests against message topics, enabling repeatable integration and L2 testing across team products.
- Developed and integrated a Java/Spring Boot aggregator service with ActiveMQ and Hermes (Node/Angular) over AMQP 1.0, supporting reliable end-to-end message flows.
- Engineered a comprehensive JUnit 5/Mockito test suite for a Spring Boot-based tactical track manager, validating partial state merges and Protobuf-to-POJO translations across Kafka and ActiveMQ transport layers.
- Wrote deep-nesting test cases for ECEF coordinates and covariance matrices to prevent null-overwriting of mission-critical track data, using Mockito static mocking and Spring's ReflectionTestUtils to isolate external dependencies.
- Refactored a Spring translation service into a modular architecture (Air, Land, Surface) and replaced an unordered HashMap with a TreeMap for relay tracking, guaranteeing deterministic ordering for IBCS aggregation and Protobuf-based health reports.

Technical Projects

4-Layer PCB – Thevenin Resistance Instrument

Jan 2026 – May 2026

- Designed and assembled a 4-layer Arduino Uno Shield in Altium with a DAC-controlled electronic load, OPAMP/MOSFET current sink, ADC voltage sensing, I2C communication, and 1:3 input divider for voltage source characterization.
- Implemented 20 pulsed current load steps at 10% duty cycle to measure voltage drop and calculate Thevenin resistance across a 12–251 mA load range.
- Developed PCB layout across two signal layers to eliminate ground-plane cross-unders, using distributed decoupling and extensive test points to improve signal integrity and board bring-up.

PCB Design – Golden Arduino

Jan 2026 – May 2026

- Designed and assembled a 2-layer ATmega328P Arduino-compatible PCB with CH340G USB-UART, ICSP programming, power switching, TVS protection, ferrite filtering, and distributed decoupling.
- Debugged initial bootloader failure by probing power and crystal signals, identifying a hidden short under the ATmega328P.
- Optimized PCB layout to minimize ground-plane cross-unders and loop area, achieving only two while maintaining accessible test points for board bring-up and signal integrity measurements.
- Characterized switching noise and transient response against a commercial Arduino Uno, measuring 4.9 ns vs. 11.0 ns 5V rise time and ~20 mV vs. ~200 mV near-field switching noise.

Embedded Software Blackjack Game System

Aug 2025 – Dec 2025

- Developed an embedded Blackjack system in C for an STM32 microcontroller with real-time LCD graphics, multi-screen UI, betting mechanics, and a cheat-code sequence using timed push-button interrupts and event-driven state transitions.

Electronics Design Lab Robot, Electrical Engineer

Jan 2025 – May 2025

- Built a Bluetooth-controlled mobile robot using an HC-05 module for wireless command transmission.
- Simulated and verified all analog subsystems in SPICE (SIMetrix), including dual DC motor models, speed sensor, compensator circuits, level-shifting comparators, RC low-pass filters, and a virtual ground reference.
- Engineered a BJT-based H-bridge to provide bidirectional control and sufficient current from a single supply.
- Programmed an interrupt-driven position control system on the Arduino Nano Every, leveraging encoder pulse feedback for closed-loop control and millimeter positional precision across both motors.